CHAPTER I

INTRODUCTION

Background of the Study

Ensuring the availability of essential medicines is a primary responsibility of public health systems and a key determinant of healthcare access and quality. Despite their importance, public health systems, particularly in low- and middle-income countries, face challenges like medicine shortages, uneven distribution, and inefficiencies in supply allocation. These issues are exacerbated by resource constraints, fragmented governance, and complex logistics systems. Studies show that even with national policies prioritizing essential medicines, their availability remains inconsistent across regions and healthcare facilities (Bazargani et al., 2015). Improvements following policy reforms are often slow and uneven, with persistent disparities limiting equitable access (Zhang et al., 2023). These findings highlight that traditional policy approaches alone are insufficient, and deeper analytical methods are needed to better understand medicine distribution dynamics.

Policy-driven interventions like centralized purchasing and restricted distribution agreements have had mixed results. Although intended to improve efficiency, these reforms sometimes fail to account for geographic dispersion, demand variability, and logistical challenges, leading to reduced delivery rates, especially in rural areas (Yang et al., 2017). This emphasizes the need to

consider both procurement policies and real-world distribution effects when assessing medicine availability.

In the Philippines, challenges are compounded by a decentralized public health system, where local governments manage procurement and resource allocation. Limited budgets and structural constraints contribute to resource scarcity, particularly at the primary care level (Bucu et al., 2023). Local practices, including informal financial arrangements, also undermine the sustainability of medicine supply systems (Collado et al., 2021), highlighting the need for a more integrated approach to healthcare resource management.

Furthermore, financial governance, budgeting, and logistics are critical in shaping medicine availability. Weak financial management and non-transparent budgeting processes often result in inequitable resource distribution and service gaps (Abidin & Tobing, 2025). Logistics issues, such as demand estimation and inventory control, also contribute to stockouts and overstocking (Priyanga & Rajalakshmi, 2024), indicating the need for an integrated system that accounts for all aspects of resource management.

Recent advances in health data analytics offer opportunities to better utilize operational data for decision-making. Health data platforms can integrate and visualize large-scale health system data, supporting policy formulation and operational planning. These analytics-driven approaches help health managers identify trends and risks, such as stockouts and supply bottlenecks (Nwokocha et al., 2025). However, the success of these approaches depends on the quality of the data and institutional readiness (Weerasinghe et al., 2022).

Unsupervised machine learning techniques, particularly clustering, have proven useful for analyzing high-dimensional and unlabeled datasets in healthcare. These methods uncover hidden patterns and anomalies, making them suitable for analyzing aggregated medicine procurement and distribution data (Wong, 2021; Eckhardt et al., 2023). Applications of clustering in healthcare resource management and operational planning support targeted and efficient resource allocation (Nnoaham & Cann, 2020; Pintakiew & Laosiritaworn, 2024). Yet, there is limited empirical work integrating unsupervised clustering with public sector medicine procurement and distribution, especially in the Philippines.

To address this limitation, this study applies two complementary clustering techniques, K-means and DBSCAN, to aggregated and non-personal medicine procurement and distribution data. K-means is a centroid-based method that partitions facilities into a defined number of groups based on similarity in procurement and distribution behavior, which supports structured comparison across facility clusters. DBSCAN is a density-based method that identifies clusters based on dense groupings in the data while also flagging outliers as noise, which is useful for detecting irregular procurement or distribution patterns that may indicate anomalies or systemic risks. Using both methods supports a more comprehensive exploration of distribution structure by capturing both compact partitions and density-driven groupings.

This study addresses this gap by applying unsupervised clustering techniques to aggregated, non-personal medicine procurement and distribution data from Philippine public health facilities. The goal is to identify hidden

distribution patterns, inefficiencies, and potential risks, contributing evidence-based insights to improve monitoring, planning, and decision-making in public health supply systems.

Objectives of the Study

General Objective

The objective of this study is to analyze aggregated and non-personal medicine procurement and distribution data from public health facilities in the Philippines using unsupervised machine learning techniques, specifically clustering analysis, in order to identify distribution patterns and potential systemic risks related to medicine availability, such as shortages, uneven distribution, or inefficiencies in supply allocation.

Specific Objectives

1. To analyze aggregated medicine procurement and distribution data from Philippine public health facilities to identify overall procurement and distribution patterns.
2. To apply unsupervised clustering algorithms, specifically K-means and DBSCAN, to group public health facilities based on similarities and variations in medicine procurement and distribution behavior.
3. To develop a web-based analytical platform that presents clustering results through cluster summaries, visualizations, and comparative profiles.

4. To evaluate and compare the clustering results of K-means and DBSCAN using appropriate validation metrics, such as silhouette score, cluster cohesion, and cluster separation.

Significance of the Study

This study is significant as it addresses critical challenges in medicine procurement and distribution within public health systems by applying data-driven analytical methods to large-scale operational data. By focusing on aggregated and non-personal medicine procurement and distribution records from public health facilities in the Philippines, the study contributes practical and methodological value to multiple stakeholders involved in healthcare delivery, policy formulation, and system management.

Public health policymakers and decision-makers, the findings of this study provide empirical evidence on distribution patterns and systemic risks related to medicine availability. The identification of clustered patterns such as uneven distribution, potential shortage risks, and inefficiencies in supply allocation can support more informed policy decisions, including the refinement of procurement strategies, allocation guidelines, and monitoring frameworks. By translating complex procurement and distribution data into interpretable patterns, the study supports evidence-based policymaking aimed at improving equity and reliability in medicine access across regions and facility types.

Health system managers and administrators, particularly those involved in logistics, supply chain management, and facility-level operations, the study offers analytical insights that can enhance operational planning and monitoring. The clustering results can help managers identify groups of facilities or medicines with similar distribution behaviors, enabling targeted interventions such as demand adjustment, inventory optimization, or focused supervision. This

contributes to more efficient use of limited public health resources and supports proactive management of medicine supply risks rather than reactive responses to shortages.

Public health facilities and frontline healthcare providers, improved understanding of medicine distribution patterns can lead to more stable medicine availability and reduced service disruptions. By identifying systemic inefficiencies and potential risks in the supply system, the study indirectly supports continuity of care, improved treatment adherence, and better health outcomes for patients who rely on public sector healthcare services, particularly in underserved and geographically isolated areas.

Researchers and scholars, this study contributes to the existing literature by integrating medicine availability research, healthcare resource management, and health data analytics through the application of unsupervised machine learning techniques. The study demonstrates the practical applicability of clustering analysis to public sector medicine procurement and distribution data, an area that remains underexplored, especially in developing country contexts. As such, it provides a methodological reference for future studies seeking to analyze complex, unlabeled health system datasets using unsupervised approaches.

In the field of *health informatics and data analytics*, the study highlights the value of unsupervised machine learning in transforming routine health system data into actionable knowledge. By showing how clustering techniques can uncover latent patterns without predefined performance labels, the study

supports the broader adoption of advanced analytics for public health monitoring and decision support.

Overall, this study is significant in advancing data-driven approaches to understanding and improving medicine availability in public health systems. By bridging gaps between operational data, analytical methods, and decision-making needs, the research contributes to strengthening public health supply systems and promoting more equitable access to essential medicines.

Scope and Limitations of the Study

Scope

This study focuses on the analysis of aggregated and non-personal medicine procurement and distribution data obtained from public health facilities in the Philippines. The scope of the research is centered on understanding patterns of medicine distribution and identifying potential systemic risks related to medicine availability through the application of unsupervised machine learning techniques, particularly clustering analysis. As part of this analytical process, the study includes the cleaning, normalizing, transforming, and structuring of the dataset to ensure data consistency and analytical readiness.

Specifically, the study covers procurement and distribution records that reflect medicine supply activities across selected public health facilities. These records may include variables such as quantities procured, delivery and distribution frequencies, stock movement indicators, and facility-level characteristics relevant to medicine supply operations. The analysis is conducted

at an aggregate level to ensure that no personal or patient-identifiable information is included.

The study is limited to the use of unsupervised analytical methods. Clustering techniques are employed to identify natural groupings and patterns within the data without relying on predefined labels or performance classifications. The primary output of the analysis consists of clustered distribution patterns that describe similarities and differences in medicine supply behavior across facilities or medicine categories.

In terms of geographic scope, the study is confined to public health facilities operating within the Philippine public health system. The findings are intended to support public health supply management, monitoring, and decision-making within this context and are not designed to represent private healthcare institutions or international supply systems.

The scope of the study is analytical and exploratory in nature. It aims to generate insights that can inform policy review, operational planning, and future research rather than to evaluate the effectiveness of specific procurement policies or to implement system-level interventions.

Limitations

Despite its contributions, the study is subject to several limitations. First, the analysis relies on secondary data derived from existing procurement and distribution records. As such, the accuracy and completeness of the findings are dependent on the quality, consistency, and timeliness of the available data. Any

missing values, reporting inconsistencies, or recording errors in the source data may influence the clustering results.

Second, the study uses aggregated and non-personal data, which limits the ability to examine patient-level outcomes, clinical appropriateness, or individual medicine utilization behaviors. The analysis focuses solely on supply-side patterns and does not directly measure health outcomes or patient access at the individual level.

Third, the use of unsupervised clustering techniques means that the identified patterns are descriptive rather than causal. While clustering can reveal similarities, anomalies, and potential risk profiles within the data, it does not establish cause-and-effect relationships between procurement practices and medicine availability outcomes.

Fourth, the study does not incorporate real-time data streams or predictive modeling. The findings reflect historical patterns present in the dataset and may not fully capture sudden disruptions, policy changes, or external shocks that occur outside the study period.

Finally, the generalizability of the results is limited to the context of Philippine public health facilities and the specific data sources analyzed. Differences in governance structures, procurement systems, and health infrastructure may limit the applicability of the findings to other countries or healthcare settings.

Definition of Terms

To ensure clarity and consistency in the interpretation of concepts used in this study, the following terms are defined operationally based on their usage within the research context:

Aggregated Data - Refers to data that have been compiled or summarized at a group or system level rather than at an individual level. In this study, aggregated data consist of combined medicine procurement and distribution records from public health facilities without identifying individual patients or personnel.

Clustering Analysis - An unsupervised machine learning technique used to group data points based on similarity across selected variables. In this study, clustering analysis is applied to identify natural groupings and patterns in medicine procurement and distribution data without predefined labels.

Data Preprocessing - The set of procedures used to prepare raw data for analysis. This includes data cleaning, normalization, aggregation, and transformation to ensure consistency, accuracy, and suitability for clustering analysis.

Distribution Patterns - Recurring trends or groupings observed in the movement, allocation, and availability of medicines across public health facilities. These patterns may indicate stable supply, irregular distribution, overstocking, or potential shortages.

Medicine Availability - The extent to which essential and non-essential medicines are accessible and sufficiently supplied within public health facilities to

meet service demands. In this study, availability is inferred from procurement and distribution data rather than direct patient-level measures.

Medicine Procurement - The process by which public health facilities acquire medicines through purchasing, ordering, or centralized supply systems. Procurement data reflect quantities ordered, sourcing practices, and supply timelines.

Medicine Distribution - The process of delivering and allocating procured medicines to public health facilities and service units. Distribution data capture the movement of medicines within the public health supply system.

Non-Personal Data - Data that do not contain personally identifiable information related to patients, healthcare workers, or individuals. This study exclusively uses non-personal data to ensure data privacy and ethical compliance.

Public Health Facilities - Government-operated healthcare institutions, such as hospitals, rural health units, and primary care centers, that provide health services within the Philippine public health system.

Systemic Risks - Potential threats to the consistent availability of medicines that arise from structural inefficiencies, uneven distribution, supply imbalances, or recurring procurement and distribution irregularities within the health system.

Unsupervised Machine Learning - A class of machine learning methods that analyze data without predefined output labels. These methods identify hidden structures or patterns within datasets. In this study, unsupervised learning is used to explore medicine supply behavior.